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Defense Expenditures, Economic Growth, and the "Peace Dividend"

A LONGITUDINAL ANALYSIS OF 103 COUNTRIES

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Despite the proliferation of studies on the impact of military spending on economic growth, it is still not known whether defense spending hinders or promotes growth. Most analysts attribute the lack of consistent/robust findings to three problems: the lack of a sound theory of defense-growth trade-offs, an inadequate research design, and a failure to account for the externality effects that defense spending might have on other parts of the economy. In this study, the authors try to remedy these problems by (1) relying on a model of the defense-growth trade-off that is grounded in the neoclassic theory of growth, (2) estimating this model using time-series data on 103 countries, and (3) explicitly accounting for the externality effects of defense spending by the use of a multisectorial model. The results show that, regardless of the specification of the models, and contrary to most cross-national studies, military expenditures have a significant positive effect on growth in only about 10% of the cases.

The question of how defense spending affects economic growth has been important to both academicians and the policy community (see Chan 1985, 1992; Fontanel 1990; Grobar and Porter 1989; Lindgren 1984, 1988; Smith 1980, 1992; Ward and Cochran 1991; Ward et al. 1991, 1992; Ward, Chan, and Davis 1993). Work in the 1970s suggested that there is a positive association between armament and development (see Benoit 1973). Likewise, the disintegration of the Soviet bloc and the apparent end to the cold war have created expectations that lower defense spending will result in a

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"peace dividend." This expectation is based on the idea that lowering military spending will lead to economic prosperity. Such an assertion, however, has had only infrequent support in the literature. In fact, despite many research efforts, no strong conclusions about the relationship between military spending and economic growth can be drawn from the literature. The inconsistent results have led Chan (1985) to conclude that a review of the literature in this area is "as likely to bewilder as it is to enlighten" (p. 405).

Several theoretical and methodological problems that have plagued research on this question may be at the root of the conflicting results (see Smith 1992). One of the most important of these problems is the lack of a strong theory underlying the empirical work in this area. Although this problem is generally recognized, few studies have attempted to resolve it. Indeed, most researchers have been content to proceed by first describing some plausible path connecting defense spending to economic performance, and then specifying ad hoc empirical models based on these conjectures. Not surprisingly, the various studies that have been generated by these efforts have differed widely in their theoretical predictions and empirical results. Some have suggested that defense spending affects economic growth positively because of human capital and infrastructure development (Benoit 1973), whereas others have emphasized possible positive effects on growth through stimulation of consumer demand (Kennedy 1983). Still others, however, have suggested that military spending crowds out civilian investment hurting overall economic performance (Smith 1980, 1992; Deger 1986; Rasler and Thompson 1988; Ward and Davis 1992; Mintz and Huang 1990, 1991).

The lack of a strong theory to guide researchers has made the formulation of empirical models particularly difficult. Saadet Deger (1986), for instance, recognized that, "in principle, economic theory should dictate what variables we should include in, for example, the growth equation." However, "in practice theory is rarely that precise . . . therefore, relatively *ad hoc* specifications are necessary" (p. 260). Given the lack of theoretically driven empirical specifications, it is little wonder that the body of empirical evidence generated on this question has been contradictory.

Much of the work in this area also suffers from limited research designs. Most studies rely on cross-sectional designs. However, the cross-sectional approach, although useful for comparative analysis, fails to capture the important dynamic element of the relationship between defense spending and the economy (Ward et al. 1991). Indeed, most recent reviews (e.g., Chan 1992) reject it as an appropriate method for testing the relationship between defense spending and growth and, instead, suggest the use of longitudinal designs.

A third problem with most of the studies that have examined the relationship between defense spending and economic growth is the failure of many

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to account explicitly for the possibility that production in the defense sector may have externality effects on production in other sectors of the economy. This, however, may be an important part of the overall effect of defense spending on the economy. For example, externality effects such as technological spillovers may be positive, whereas effects such as the diversion of economic inputs from the civilian sector may cause negative externalities.

When one considers the above problems, it is not surprising that the literature has failed to provide any meaningful consensus on the question of whether defense spending encourages or hinders economic growth. Indeed, any study that fails to address these issues is unlikely to contribute to such an answer. This study, therefore, tests for the impact of defense spending on growth by using (1) a model of growth that is grounded soundly in economic theory, (2) individual longitudinal analyses of 103 countries, and (3) an externality component incorporated into the theory of growth. In so doing, this analysis is unique—because most other work has either examined many countries in cross-sectional analysis (e.g., Smith 1980; Biswas and Ram 1986) or a few countries in time-series (longitudinal) analysis (e.g., Chan 1988; Davis 1991; Ward and Cochran 1991; Ward and Davis 1992; Ward et al. 1991, 1992).

To accomplish this, we first review the logic of the incorporation of defense-spending variables into the neoclassic model of economic growth (see Solow 1988; Denison 1967, 1985; Feder 1982; Ram 1986, 1989). Then, extending this model, we undertake an empirical analysis of the impact of defense spending on growth in 103 countries over time.

THE MODEL

Recent studies of the defense-growth trade-off (e.g., Deger 1986; Fontanel 1990; Smith 1992; Ward and Cochran 1991; Ward et al. 1991) represent only the latest round of an ongoing debate about whether defense spending actually hinders growth. On one side of the argument is work that relies on Marxist writers to suggest that military expenditures are used to increase the profitability of monopoly sector industries (Baran and Sweezy 1966; O'Connor 1973; Griffin, Devine, and Wallace 1982). On the other side of the argument are those researchers who posit a negative relationship between defense spending and growth (Deger 1986; Rasler and Thompson 1988).

The model developed below is an extension of a model that first appeared in Feder (1982) and was later used in an influential article by Rati Ram (1986). In this article, Ram used the "augmented neoclassic model" to claim,

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on the basis of cross-sectional and time-series data, that government size has a positive impact on economic growth. Drawing on Denison's (1967, 1985) sources-of-growth model that relates changes in levels of gross output to changes in inputs of capital and labor, Ram (1986) proposed a two-sector model of the economy consisting of the civilian and government sectors.

Ram (1986) claimed that the inclusion of the government sector in the growth model is justified because government could affect growth negatively because of inefficient operations, regulatory costs, and distortions of normal market incentives caused by governmental policy. Government could also influence growth positively, however, through its role in "settling conflicts between private and social interests," the prevention of foreign economic exploitation, the encouragment of productive investment, and the provision of a socially optimal direction for growth and development (Ram 1986). Perhaps surprisingly, Ram found that in most countries the government affects growth positively. Ram's findings thus provide justification for the inclusion of government as an important component of economic growth. However, in Ram's model, the military sector is only considered insofar as it is part of the overall government sector, and thus the model implicitly assumes that the effects of military production are indistinguishable from those of the government sector in general. Thus Ram's original model fails to account for any independent effect that the military sector may have on economic growth (see Carr 1989; Rao 1989).

In the context of the present study, Ram's study is important because it implies that all government expenditures (both nonmilitary and military) have a positive impact on economic growth. As shall be demonstrated below, however, this conclusion does not necessarily hold when Ram's model is expanded to account for the separate effects of both military and nonmilitary government spending.

Using a supply-side or production function approach is not new to defense studies. Indeed, a production function relating output to capital, labor, and military spending has been used in several studies. This specification allows for the straightforward estimation of the marginal product of military inputs on output. However, it also relies on the assumption that the economy represented by the production function is homogeneous. That is, there can be no difference between the kinds of factor inputs in the military and other sectors in the economy, or in the factor productivities of particular inputs between sectors.

Nevertheless, there are many reasons to suspect that the military sector is different from other sectors in the economy. For example, it is not at all clear that the incentives underlying the decisions of management in military firms are the same as the incentives facing their civilian counterparts. This has led

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some scholars to suggest that the productivity of factor inputs such as capital and labor may be quite different in the military and civilian sectors of the economy (e.g., Dumas 1976).

Mintz and Huang (1990, 1991) extended Ram's model to encompass a three-sector economy. Noting that the government sector consists of both military and nonmilitary components, Mintz and Huang (1990) included both as separate sectors and estimated an equation in which the government, military, and civilian sectors enter separately (see Ward et al. 1991; Ward and Cochran 1991; Davis 1991; DeRouen 1991, 1992; and McMillan 1992 for tests of variants of this model with data on Argentina, Brazil, India, South Africa, and other developing countries).

Another phase in the development of this model followed an extension of the theoretical model by Huang and Mintz (1991) to account not only for the overall effects of the military sector on economic growth but also for the externality effects of military spending (for a discussion of the concept of externality, see Cornes and Sandler 1986). This model is specified below.

SPECIFICATION OF THE MODEL

Ram's (1986) growth equation is

$$\dot{Y}/Y = \alpha(I/Y) + \beta(\dot{L}/L) + (\delta' + C_G)[(\dot{G}/G)\,(G/Y)], \tag{1}$$

where a dot over the variable indicates its differentiation with respect to time, the parameter α is the marginal product of capital in the C sector, $I = dk/dt$ is investment, and β reflects the linear relation between marginal labor productivity in the C sector (C_L) and average output per laborer (Y/L) (see Huang and Mintz 1991). The sum of δ' and C_G is interpreted as the overall effect of the government sector on economic performance, and δ' equals $\delta/(1 + \delta)$ and is the relative productivity effect of the government on economic growth. C_G, on the other hand, represents the marginal externality effect of the government sector on the nongovernment sector (Huang and Mintz 1991).

To isolate the effect of military expenditures on economic growth, Mintz and Huang (1990) have disaggregated government output, measured by total government purchases of goods and services, into nonmilitary and military components ($G = N + M$). Assuming that both N and M are separate sectors in the economy, assigning them separate production functions with inputs of labor and capital in each sector, and rederiving equation 1 for three sectors, Mintz and Huang (1991) obtained the following statistical equation:

$$\dot{Y}/Y = \alpha(I/Y) + \beta(\dot{L}/L) + (\delta'_n + C_N)\,[(\dot{N}/N)(N/Y)] + (\delta'_m + C_M)\,[(\dot{M}/M)(M/Y)], \tag{2}$$

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where ($\delta'_i + C_i$, $i = n, m$) is interpreted as the overall effect of sector i on economic performance. That is, it captures both the direct effect of sector i on growth and the effect felt through the interaction of sector i with other sectors.

To identify externalities in equation 2, Huang and Mintz (1991) have further assumed that N and M affect the production of C with constant elasticities of π_n and π_m, respectively (see Feder 1982; Ram 1986, 1989):

$$C = C(L_C, K_C, N, M) = N^{\pi n} \bullet M^{\pi m} \bullet \phi\,(L_C, K_C).$$

One can then show that

$$\partial C/\partial N = C_N = \pi_n \bullet (C/N),$$

$$\partial C/\partial M = C_M = \pi_m \bullet (C/M).$$

Using the above results in equation 2 results in

$$\begin{aligned} \dot{Y}/Y = \alpha(I/Y) + \beta(\dot{L}/L) + [\delta'_n + \pi_n \bullet (C/N)] \\ [(\dot{N}/N)\,(N/Y)] + [\delta'_m + \pi_m \bullet (C/M)]\,[(\dot{M}/M)\,(M/Y)]. \end{aligned} \tag{3}$$

By rearranging the terms, this equation becomes (Huang and Mintz 1991):

$$\begin{aligned} \dot{Y}/Y = \alpha(I/Y) + \beta(\dot{L}/L) + \delta'_n\,[(\dot{N}/N)\,(N/Y)] + \\ \pi_n\,[(\dot{N}/N)\,(C/Y)] + \delta'_m\,[(\dot{M}/M)\,(M/Y)] + \pi_m\,[(\dot{M}/M)\,(C/Y)]. \end{aligned} \tag{4}$$

Intuitively, the elasticity measure of the intersectoral externality π_i ($i = n$, m) may be interpreted as the effect on economic growth of the interaction between the growth rate of sector i and the share of the nongovernment sector (C) in total output (Y). Likewise, δ'_i can be interpreted as above so that δ'_i captures the direct effect of sector i on growth, whereas π_i separates out the "externality effect," that is, the effect of sector i on growth through its effects on other sectors. Thus the equation allows for the isolation of both the productivity *and* externality effects of military and nonmilitary spending (Huang and Mintz 1991).

DATA AND METHOD

Equations 2 and 4 are estimated for each of the 103 countries in the sample using Summers and Heston's (1988) economic data (also known as the

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International Comparisons Project or ICP data set) and the Stockholm International Peace Research Institute's (SIPRI) military expenditures data.

Specifically, investment, government consumption, population, and output have been taken from the Summers and Heston data set. ICP economic data (in 1980 international prices) have been widely used in studies of growth (see, e.g., Ram 1986). The SIPRI data on military spending were collected in constant dollars for various years and converted to a 1980 base. The countries included in the sample, as well as the time periods for which data were available, are given in the appendix.

Summers and Heston (1988) ranked the quality of the economic data for each of the countries in their sample. Likewise, SIPRI (1980) has provided a ranking of the military expenditure data for most of the countries in the world on the basis of the degree to which standard sources differ. These rankings are provided in the appendix.

The use of the ICP economic data has both advantages and disadvantages. On the positive side is the fact that these data are the most reliable data on output and its components currently available. Further, the extensive coverage makes it possible to evaluate the model in over 100 different countries, thus our conclusions can be based on general trends and not on the idiosyncracies of a few cases. Furthermore, Summers and Heston's (1988) evaluation of the relative quality of the data allows us to assess the robustness of our findings across subgroups of countries with data of similar quality.

On the other hand, these data were not collected with our model in mind. Thus they do not fit our data requirements perfectly. Specifically, one of the variables used in our analysis is nonmilitary government spending. One approach to deriving the values for this variable from the available data is to subtract the SIPRI military expenditure data from the Summers and Heston government consumption data. This approach, however, has the drawback that when military capital expenditure (which is not included in the government consumption data) is large, the derived values for nonmilitary government spending may be artificially low.[1]

Alexander (1990) took another approach to the problem of measuring nonmilitary government spending. That is, he did not subtract military spending from government consumption; instead, he used government consumption as a proxy for nonmilitary government consumption (thus over-

1. Looking at a few cases where data on both military consumption and military expenditure are available revealed few significant differences (the countries compared were Algeria, Australia, Austria, Belgium, Benin, Bolivia, Fiji, Gabon, Greece, and India). On the other hand, the Middle Eastern countries appear to have systematically large differences between military expenditure and military consumption. Because of this problem, we consider all of our results both including and excluding the Middle Eastern cases.

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valuing government consumption by the amount of military consumption). The discussion below focuses on the results of the analysis using the first method. However, estimates using Alexander's (1990) method were also obtained and are reported for purposes of comparison.[2]

Following Ram (1986) and others (Ward et al. 1992; Lebovic and Ishaq 1987; Alexander 1990), this study uses the rate of population growth in place of the rate of increase in labor. As Ram (1986) pointed out, although this is not a good proxy in some cases, it has the advantage that data on population tend to be reliable, whereas time-series data on labor force are rare and of questionable quality, particularly for the less-developed countries. Further, Lebovic and Ishaq (1987, 118) have suggested that, because labor participation rates show little volatility in the short run, the population growth rate may be used instead. Results based on this proxy variable should be viewed with care, however.

THE MILITARY-EXPENDITURE DATA

Many scholars have criticized the quality of the military-expenditure data on which most studies in this area rely, suggesting that they may not be comparable across countries (see Brzoska 1981). All three major sources of military-expenditure data (SIPRI, U.S. Arms Control and Disarmament Agency [ACDA], and International Institute for Strategic Studies [IISS]) rely, at least in part, on definitions of military expenditure that are different for different countries or groups of countries. Thus the comparability of data from countries using different definitions is highly questionable. Nevertheless, the present analysis does not require direct comparison of data *across* countries, but only comparison of results derived from individual time series. The cross-national comparability problem then is not severe in this case (see Chan 1985 on this point).

Because of the above problem, and other problems more relevant to longitudinal designs (e.g., cases where the data collection procedures changed over the years in the data source), Chan (1985) and others have suggested that multiple data sources for military spending should be used (e.g., SIPRI and the ACDA data set). Others (notably Ward et al. 1991) have used data obtained from official country-specific sources (i.e., government publications). Although the task of conducting such an analysis on 103 countries over time is beyond the scope of this article, we do not ignore

2. It should be noted that our analysis uses different data sources and has longer time series than Alexander's (1990) nine-country study.

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problems of data quality. As mentioned above, one way in which we try to minimize problems of data quality is to consider our results separately for different groups of countries with data of similar quality. For the military-spending data, the criteria used by SIPRI's scholars for evaluating the reliability of the data is the degree to which the major sources of data on military spending (i.e., SIPRI, ACDA, IISS) converge for a given country. Thus consideration of the subset of countries where convergence is high gives us some confidence that the use of alternative data sources, at least for these countries, would likely lead to results similar to ours.

Finally, a brief discussion of the comparability of military-spending data and the Summers and Heston ICP data is in order. In this study we use SIPRI's military expenditure data, which were converted from domestic currencies by means of exchange rates. The use of these data with the ICP estimates is justifiable because ICP estimates of the value of internationally traded goods (such as weapon systems and military technology) tend to be very close to those valued using exchange rates (see Hogendorn 1987, 35-37 on this point). The advantages provided by the spatial and temporal coverage of the data, as well as the superior reliability of the ICP economic data, outweigh, in our opinion, the relatively minor differences in the valuing of the two series.

ESTIMATION

Equations 2 and 4 were estimated using the ordinary least squares (OLS) procedure, except where the existence of autocorrelation or multicollinearity in the data required the use of an alternative estimation procedure.[3] When the presence of autocorrelation was indicated by the Durbin-Watson test statistic, the generalized least squares procedure was used instead. When the Variance Inflation Factor (VIF) statistic indicated that the independent variables were collinear, estimates were also obtained using the ridge regression technique.

The time-series properties of the data were investigated using standard procedures. Specifically, Dickey-Fuller tests accounting for both drift and a deterministic time trend were used to test the series for stationarity, whereas a one-step forecast F test and the Cumulative Sum of Residuals (CUSUM) test were used to test for parameter instability. The results of the Dickey-Fuller tests (available from the authors) for the variables used in equations 2

3. Chowdhury (1991), Deger (1986), and others have explored different forms of causality analysis and have convincingly showed that economic growth does not significantly influence military spending in most developing countries. Thus the direction of the causality problem raised by Rao (1989, 278) should not concern us here.

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and 4 revealed the existence of a unit root and, hence, nonstationarity (at $p <$ .05) in the economic-growth series for only two cases (Philippines and Guatemala). In addition, the existence of a unit root in the military and nonmilitary externality variables was rejected in every case. Further, we could not reject the existence of a unit root for the nonmilitary-spending series for only one country (Tanzania), and for the military-spending series we could not reject the existence of a unit root for New Zealand and Chad. Given the well-known bias of these unit root tests toward not rejecting the null hypothesis of a unit root, we also considered the results using a significance level of .10. When this criterion is used, no unit roots are found in any of the military and nonmilitary series that appear in equations 2 and 4. In general, these results suggest that nonstationarity of the variables in the equations is not a problem for this study.[4]

Tests for parameter instability were also conducted for all the equations for each country. These tests are based on the recursive residuals obtained from the estimation of the equations (see Green 1993, chap. 7). The tests indicated that, for the vast majority of cases in the sample, parameter instability was not a problem. On the other hand, there were 16 countries in which tests of equation 2 suggested that the relationships modeled may have changed in the mid- to late 1970s. In general, the timing of this structural break for these countries corresponds to the second oil shock; however, there seems to be no pattern with regard to the particular countries that show significant effects.[5]

To further enhance the validity of the study, we report two sets of findings. The first provides estimates (for each of the 103-country sample) of the size

4. This conclusion may be modified somewhat given the results of the unit root tests for the other variables in equations 2 and 4, that is, labor growth and investment. For these variables, nonstationarity in the time series is more common. At a .05 significance level, 56 cases failed to reject a null of a unit root for investment and 27 failed to reject for labor growth. At a .10 significance level, these figures are 39 and 18, respectively. The existence of nonstationary variables on the right-hand side when the dependent variable is stationary may have two effects. First, the coefficients on the nonstationary variables should theoretically be zero in infinite samples. However, our finite sample results do not conform to this theoretical expectation. Second, the existence of nonstationary variables on the right-hand side may lead to inefficiencies that affect all the variables in the equation. To the extent that this occurs, then, it acts to undermine our essentially negative conclusions. On the other hand, an examination of the 26 ($p < .05$) or 56 ($p < .10$) cases that have no unit root on the right-hand side continues to support our main conclusion.

5. The countries that showed evidence of parameter instability for equation 2 were Belgium, Chile, France, Gabon, Guyana, India, Japan, Kuwait, Luxembourg, the Netherlands, Niger, Portugal, South Africa, S. Korea, Switzerland, and Trinidad and Tobago. For equation 4, the above countries continued to show evidence of parameter instability and three other countries—Algeria, Hong Kong, and former West Germany—did as well.

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and externality effects of military spending on growth using Ward et al.'s (1991) and Mintz and Huang's (1990, 1991) method for measuring nonmilitary spending. The second set provides estimates of the same effects (for the same sample) using Alexander's (1990) method of calculating nonmilitary government spending. In addition, the results of the analysis are considered for various subsamples of countries for which data reliability varies (see below).

RESULTS

We first present the results for the total effect of military spending on growth (equation 2) and then consider externality effects (equation 4).[6]

Table 1 reports the coefficients for the military and nonmilitary government-spending variables in equation 2 for each of the 103 countries in the sample. The first thing to note about these results is that they lend support to Ram's (1986) conclusion that government spending is an important positive influence on economic growth (or at least that it is not a negative influence).[7] Indeed, the effect of nonmilitary government spending is significant and positive in about 40% of the cases, whereas it is negative in only

6. Looking at the R^2 for equation 2 for each of the 103 time series, we find that the mean value of R^2 is .44. Likewise, for equation 4 this figure is .23. In general, the range of the values for the R^2 was small for equation 2, whereas for equation 4 the mean R^2 reflects a number of very low values that tended to pull down the average. Overall, these figures suggest that these models capture a significant portion of the process driving economic growth in the countries in the sample in equation 2 (and less so in equation 4). Thus they provide an appropriate context in which to evaluate the effects of military and nonmilitary government spending.

7. Another consideration when evaluating the overall performance of the model concerns the coefficients on the investment and labor variables. In general, we will be more confident in the specification of the model if the coefficients on these variables conform to the standard predictions of economic theory. For instance, if we found investment to be a consistently negative influence on growth, we might question the validity of the model.

This, however, is not the case. With regard to the coefficients on the investment variable, the models perform well. For equation 2, there is only 1 case in which investment is related negatively to economic growth (Saudi Arabia), whereas the relationship is positive and significant in 43 ($p < .05$) or 48 cases ($p < .10$). For equation 4, investment is significant and positive in 44 ($p < .05$) or 50 ($p < .10$) cases and negative in only 2.

The economic theory underlying the impact of labor on the economy is less conclusive than that of investment; however, the general prediction is that labor growth should lead to economic growth. Nevertheless, our results are mixed, with population growth in equation 4 significant at the .05 level in 17 cases and at the .10 level in 25 cases (4 of these being negative at .05 and 7 at .10). In equation 4, 13 ($p < .05$) or 16 ($p < .10$) cases have positive coefficients and 9 ($p < .05$) or 10 ($p < .10$) cases are negative. Of course, the use of population as a proxy for labor may cause the impact of labor to be underestimated, especially in cases where the size of the labor force changed significantly while population remained stable.

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TABLE 1
Estimates of Equation 2: The Effects of Nonmilitary and Military Spending on Economic Growth

Statistical Significance of the Estimated Coefficients	*Nonmilitary Spending*		*Military Spending*	
	Country	*Coefficient*	*Country*	*Coefficient*
Significant at .05				
	Bolivia	0.78	Colombia	1.03
	Burma	1.77	Ecuador	34.10
	Burundi	1.66	Ethiopia	3.00
	Chile	2.05	Iran	2.14
	Cyprus	3.95	Iraq	4.44
	Dominican Republic	1.35	Ireland	2.30
	El Salvador	1.10	Israel	0.59
	Fiji	2.32	Sierra Leone	23.20
	Guatemala	1.52	Spain	10.10
	Guyana	0.21	Taiwan	5.78
	India	0.77	Uganda	2.54
	Iran	2.14		
	Iraq	4.98		
	Ireland	2.12		
	Israel	0.34	Argentina	–6.02
	Italy	2.62	Australia	–2.61
	Kenya	1.20		
	Korea, South	2.79		
	Madagascar	1.72		
	Malawi	1.05		
	Malaysia	3.78		
	Mali	1.59		
	Mexico	3.33		
	Netherlands	2.39		
	Nicaragua	1.38		
	Niger	4.59		
	Norway	2.55		
	Oman	0.65		
	Pakistan	1.29		
	Philippines	1.34		
	Rwanda	3.22		
	Senegal	2.12		
	Spain	7.10		
	Sri Lanka	1.95		
	Syria	3.56		
	Taiwan	4.48		
	Trinidad and Tobago	2.05		

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TABLE 1 continued

Statistical Significance of the Estimated Coefficients	*Nonmilitary Spending*		*Military Spending*	
	Country	*Coefficient*	*Country*	*Coefficient*
	Uganda	1.86		
	Zimbabwe	1.24		
	Australia	–3.38		
	Finland	–4.80		
Not significant at .05	All other countries in the sample; 63 cases (see appendix)		All other countries in the sample; 91 cases (see appendix)	

two cases. These results for nonmilitary government spending are important because they can help us understand the relative impact of military and nonmilitary spending and thus, to some extent, assess the opportunity cost of using government resources for military purposes.

This conclusion, however, does not hold up when one considers the military component of government spending. As the second column of Table 1 shows, the overall effect of military spending on economic growth is not statistically different from zero in the vast majority of cases (about 90%). These results suggest then that there is no short-term effect of military spending on economic growth when considering the overall impact.

Equation 2, however, is only part of the story. It is possible that government spending affects economic growth through its effect on other sectors of the economy. An investigation of these effects is important if we desire to understand the underlying mechanisms by which defense spending affects economic growth. For example, if we should find significant externality effects, even though the overall effect of defense spending on growth is negligible, then we might conclude that other factors must be cancelling out any externality effect. The possibility of such "externality" effects is accounted for in equation 4, and the results of the estimation of this equation are given in Table 2. As is clear from the table, the explicit incorporation of externality effects into the model does little to change the conclusions drawn from equation 2, that is, that nongovernment spending often has positive effects on the economy, whereas military spending has no significant effect.[8] The results in Table 2 also reveal another interesting pattern; that is, the

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"externality" effect for both nonmilitary and military spending is small in relation to the overall effect of these variables. Comparing the coefficients in the two tables, it is apparent that the total effects are typically several times larger than the "externality" effects.

To enhance the validity of the results, we also estimated equations 2 and 4 using Alexander's (1990) method for calculating nonmilitary government spending. The results from this estimation support the above findings. Specifically, the estimates for the nonmilitary-spending variables in both equations are very similar to those reported in Tables 1 and 2. Further, estimates of equation 2 produced only 5 significant coefficients (at the .05 level) for the military-spending variable (2 positive and 3 negative), whereas estimates of equation 4 produced only 10 significant coefficients on the military-spending variable (8 positive and 2 negative).

It is also not the case that all the data used in the analysis are of equal quality. One concern then is that our results could be related in some way to these differences. Our analysis shows, however, that this is not the case. A comparison of both Summers and Heston's (1988) and SIPRI's (1980) ranking of the quality of the economic- and military-expenditures data (see appendix) with the results in Tables 1 and 2 show that there is no systematic relationship between the relative quality of the data series used in the analysis of a given country and the size or significance of the coefficients for that country. Further, restricting our attention to the cases in which we have the most reliable data (rated A or B on the economic variables and A on military spending), we come to the same conclusions as when looking at the whole sample. That is, out of 21 cases only 1 is significant for military spending in equation 2 and only 3 in equation 4.

In addition, for some countries in Tables 1 and 2 (those marked with an asterisk in the appendix), the measurement problems associated with non-

8. The military and nonmilitary spending and externality variables in equation 4 were multicollinear for nearly every country in the sample. Because multicollinearity tends to bias hypothesis tests toward failing to reject the null, and because our primary conclusion is that there are no statistically significant effects for the military variables in equation 4, it is possible that conclusions based only on this equation are an artifact of the collinearity between the variables in the equation. To avoid this problem, the equation was also estimated using the ridge regression technique. This technique introduces small biases in the coefficient estimates to achieve substantial gains in variance reduction. The ridge regression coefficients for equation 4 are the ones that appear in Table 2. As expected, the results obtained when using the ridge technique revealed many more significant coefficients for all the variables in the model. For instance, the ordinary least squares results revealed only 20 significant coefficients for the nonmilitary externality effect, whereas the ridge estimates revealed 42. The sizes of the coefficients, on the other hand, were relatively stable between ridge and nonridge estimates despite the introduction of some bias into the ridge estimates. The coefficients in Tables 1 and 2 thus remain comparable.

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TABLE 2
Estimates of Equation 4: The Externality Effects of Nonmilitary and Military Spending on Economic Growth

Statistical Significance of the Estimated Coefficients	*Nonmilitary Spending*		*Military Spending*	
	Country	*Coefficient*	*Country*	*Coefficient*
Significant at .05				
	Argentina	0.27	Austria	0.02
	Belgium	1.39	Congo	0.27
	Colombia	0.10	West Germany	0.09
	Denmark	0.37	Guatemala	0.03
	El Salvador	0.26	Morocco	0.34
	France	0.67	Pakistan	0.17
	India	0.78	Singapore	0.11
	Ireland	0.33		
	Israel	0.16		
	Jamaica	0.66		
	Japan	0.39		
	Jordan	2.27		
	Korea, S.	0.26	Canada	–1.04
	Madagascar	0.21	Nicaragua	–0.02
	Malaysia	0.49		
	Mexico	0.51		
	New Zealand	2.26		
	Niger	0.04		
	Nigeria	0.11		
	Pakistan	0.19		
	Peru	0.55		
	Philippines	0.74		
	Rwanda	0.57		
	Somalia	0.39		
	South Africa	0.69		
	Spain	1.08		
	Sri Lanka	0.26		
	Sweden	1.08		
	Syria	1.03		
	Taiwan	0.79		
	Thailand	1.00		
	Tunisia	1.88		
	Zambia	0.48		
	Australia	–0.24		
	Bolivia	–0.17		
	Congo	–1.24		
	Cyprus	–3.15		

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TABLE 2 continued

Statistical Significance of the Estimated Coefficients	*Nonmilitary Spending*		*Military Spending*	
	Country	*Coefficient*	*Country*	*Coefficient*
	Italy	−0.23		
	Netherlands	−0.80		
	Uganda	−0.54		
	United States	−0.57		
	West Germany	−0.86		
Not significant at .05				
	All other countries in the sample; 62 cases (see appendix)		All other countries in the sample; 95 cases (see appendix)	

military government spending are particularly problematic. Looking at the tables, however, we see that excluding these 10 Middle Eastern cases from consideration acts only to reinforce our conclusion by further reducing the number of countries in which there is a statistically significant positive relationship between military spending and economic growth.

The lack of significant short-term negative effects of military spending on growth should not come as a surprise to students of defense politics and economics. After all, most scholars agree that it is the *long-term* effect of military spending on development that is negative, whereas the *short-term* effects are thought to be positive (for reviews, see Lindgren 1984; Chan 1991, 1992).

The major contribution of this study is the finding that there is no strong short-term significant relationship between military spending and the economy, even when looking at the externality effects separately. Indeed, this result, coming as it does from an analysis of more than 100 individual time series is the most consistent answer to Benoit's (1973) finding of a possible positive relationship between armament and development. These findings also have important implications for the debate over the "peace dividend" in that they suggest that the economic benefits of reductions in military spending may be felt only in the long run.

Finally, a natural question to ask when interpreting these results is whether or not the countries that fall into the various cells in Tables 1 and 2 conform to any patterns that would provide clues to the structural factors that affect the relationship between government spending and growth. For instance, on

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the basis of a cross-sectional analysis, Looney (1988) has suggested that greater arms production capacity and resource availability leads to a positive relationship between military spending and economic growth in Third World countries. Our results may lend some support to this view, in that Israel, Taiwan, Iran, and Iraq are among the few countries with positive coefficients for the overall effect of military spending. On the other hand, other Third World producers such as South Korea and Brazil have insignificant coefficients, whereas Argentina has a significantly negative coefficient.

With regard to other factors that may affect the relationship between military spending and growth, it is interesting to note that many of the countries that have positive effects are also countries that have been involved in protracted conflicts during the period under study and have received very large amounts of military aid. Indeed, 7 of the 11 countries that have positive coefficients have experienced significant external security threats and have received massive military aid from the superpowers[9].

Despite the robustness of the results, however, the question of the true relationship between defense spending and economic growth remains only partially answered. Our results do not rule out the possibility that the long-term effects of military spending are indeed negative, that the relationship between military spending and economic growth is nonlinear (see Smith 1992, 14), or that there may be unexpected consequences of the "peace dividend" in the form of increased savings because of reduced fear of nuclear war (Russett and Slemrod 1992). Also, military spending may affect growth indirectly via, for example, investment (the crowding-out effect; see Smith 1980), labor (see Dunne and Smith 1990), or the deficit.

CONCLUSIONS

A crucial part of the logic underlying the "peace dividend" is the idea that reductions in military spending lead to growth. On the other hand, Ram's (1986) influential study implies that both civilian and military expenditures encourage growth—results that, if widely accepted, would encourage governments to promote military programs and might also discourage governments from pursuing policies aimed at reducing defense spending. Our estimates of a model that is grounded soundly in economic theory, addresses externality effects, and uses longitudinal data from 103 countries showed that, although nonmilitary spending has a significant and positive impact on

9. At a .05 significance level and 103 countries, one would expect approximately five cases in which we reject the null hypothesis of no relationship when, in fact, no relationship exists.

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growth in many countries, the impact of military spending on growth is largely insignificant. Therefore, governments should not be encouraged to expand military programs for economic purposes.

These results may also speak to recent developments in World Bank and International Monetary Fund policy that explicitly discourage military spending in countries expecting development assistance. Previous studies (e.g., Benoit 1973) suggest that defense spending encourages growth. Our results show, however, that increases in *non*military spending contribute to growth significantly more than increases in military expenditures.

One of the most important consequences of the end of the cold war is the lessening of the superpower arms race and its spillover effect to other geographic areas. The frequent use of the term *peace dividend* suggests that many people believe there is a relationship between reduced defense spending and increased economic performance. Although the liklihood and extent of any peace dividend will depend ultimately on how much, how soon, and what kind of military spending will be cut as well as onto what alternative uses these resources will be put (Mintz 1992), it is clear that, when the threat of a superpower war has decreased, many have come to expect defense burdens to shrink and there to be a resultant positive effect on economic performance.

Kennedy (1987), Rasler and Thompson (1988), and others have suggested that heavy defense commitments and high defense budgets cause dominant nations to decline economically and, thus, to loose their leading position in the international system. Others (e.g., Smith 1980) suggested that high defense budgets indirectly undermine economic performance. Both the link between global security and defense spending and that between defense spending and economic performance are at the core of the "peace dividend" debate and are vitally important in characterizing the way in which global changes will affect economic performance. If, for instance, the end of the cold war and trends toward democratization bring substantial increases in global security, and if this results in reductions in defense spending, yet there exists only an indirect relationship between defense spending and economic performance as most scholars have suggested, then the "peace dividend" may only be felt in the long run.

This study confirms that the peace dividend, if it comes at all, will likely not be felt in the short run. Further, because this research has relied on a sample of 103 countries over time, has incorporated both cross-sectional and longitudinal analyses, and has taken into account externality as well as productivity effects, it provides strong, systematic evidence for a result that has previously relied on the analysis of individual case studies.

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APPENDIX

Country	*Coverage*	*Economic Data*	*Military Spending*
Algeria	62-85	C	C
Argentina	59-85	B	C
Austria	50-85	A	A
Australia	50-85	A	B
Bangladesh	71-85	C	C
Belgium	50-85	A	A
Benin	63-85	B	—
Bolivia	56-85	B	B
Brazil	56-85	B	C
Burma	50-85	C	B
Burundi	64-85	D	—
Cameroon	60-85	C	B
Canada	50-85	A	A
Central African Republic	60-85	D	—
Chad	67-85	D	B
Chile	57-85	C	C
Colombia	50-85	B	B
Congo	63-85	D	C
Costa Rica	50-85	B	—
Cyprus	66-85	B	—
Denmark	50-85	A	A
Dominican Republic	50-85	C	—
Egypt*	50-85	D	C
El Salvador	53-85	B	B
Ecuador	55-85	B	B
Ethiopia	59-85	C	B
Fiji	68-85	C	—
Finland	50-85	A	A
France	50-85	A	A
Gabon	64-85	C	—
West Germany	50-85	A	A
Ghana	59-85	D	B
Greece	50-85	A	A
Guatemala	50-85	B	A
Guyana	68-85	C	—
Haiti	60-85	C	—
Honduras	50-85	C	B
Hong Kong	62-85	A	—
India	50-85	B	B

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APPENDIX continued

Country	*Coverage*	*Economic Data*	*Military Spending*
Indonesia	62-85	B	B
Iran*	57-85	C	C
Iraq*	53-85	C	C
Ireland	50-85	A	B
Israel*	50-85	A	C
Italy	50-85	A	D
Jamaica	64-85	C	—
Japan	51-85	A	A
Jordan	54-85	C	B
Kenya	56-85	B	B
Korea, South	53-85	B	A
Kuwait*	61-85	B	C
Liberia	65-85	D	—
Luxembourg	50-85	A	A
Madagascar	62-85	C	A
Malawi	66-85	C	—
Malaysia	55-85	B	B
Mali	63-85	C	—
Mauritania	61-85	D	—
Mauritius	60-85	D	—
Mexico	50-85	B	B
Morocco	58-85	C	B
Nepal	60-85	D	B
Netherlands	50-85	A	A
New Zealand	74-85	A	A
Nicaragua	64-85	C	A
Niger	63-85	D	—
Nigeria	58-85	C	C
Norway	50-85	A	A
Oman*	70-85	B	C
Panama	64-85	B	—
Pakistan	50-85	B	B
Paraguay	62-85	C	B
Peru	52-85	C	B
Philippines	52-85	A	C
Portugal	50-85	A	B
Rwanda	65-85	D	—
Saudi Arabia*	61-85	C	C
Senegal	61-85	C	A
Sierra Leone	62-85	D	—

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APPENDIX continued

Country	*Coverage*	*Economic Data*	*Military Spending*
Singapore	60-85	C	C
Somalia	63-85	D	A
South Africa	50-85	B	B
Spain	50-85	A	C
Sri Lanka	58-85	B	B
Sudan	55-85	D	B
Sweden	50-85	A	A
Switzerland	50-85	B	A
Syria*	60-85	C	B
Taiwan	53-85	B	A
Tanzania	64-85	C	B
Thailand	50-85	C	A
Togo	61-85	D	—
Trinidad & Tobago	60-85	C	—
Tunisia	60-85	C	D
Turkey	50-85	B	B
Uganda	60-85	D	C
United Arab Emirate*	70-85	C	C
United Kingdom	50-85	A	A
United States	50-85	A	A
Uruguay	61-85	B	B
Venezuela	50-85	B	A
Zaire	65-85	D	B
Zambia	55-85	B	B
Zimbabwe	54-85	C	C

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